

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q20-1397

Norfolk 138 kV  
July 25, 2021

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1397. The Interconnection Customer proposes a 123.1 MW natural gas combined cycle facility interconnecting at the Norfolk 138 kV substation in the Virginia area. The request was received on January 29, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 444 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$655,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

1. Facility: Rocky Reach – Idaho Springs 138.0 kV  
Contingency: Sideburn – Yavaai 138.0 kV  
Loading: 110.7% (Rating: 255 MVA)  
DFAX: 9.5%  
Pre-project loading: 106.2%
2. Facility: Centennial – Glen Carlyn 138.0 kV  
Contingency: Gallows Rd – Idylwood 138.0 kV  
Loading: 108.1% (Rating: 278 MVA)  
DFAX: 5.1%  
Pre-project loading: 105.8%
3. Facility: Rocky Reach – Burke-Sideburn 138.0 kV Line  
Contingency: Sideburn – Yavapai 138.0 kV  
Loading: 107.9% (Rating: 238 MVA)  
DFAX: 9.5%  
Pre-project loading: 103.0%
4. Facility: Annndale – Centennial 138.0 kV Line  
Contingency: Gallows Rd – Idylwood 138.0 kV L  
Loading: 105.3% (Rating: 345 MVA)  
DFAX: 5.1%  
Pre-project loading: 103.4%
5. Facility: Gallows Road – Idylwood 138.0 kV line  
Contingency: Annandale – Centennial 138.0 kV  
Loading: 105.2% (Rating: 323 MVA)  
DFAX: 5.6%  
Pre-project loading: 103.1%

6. Facility: Covanta Fairfax Energy – Occoquan 138.0 kV  
 Contingency: Fairfax – Jeune Landing 138.0 kV L  
 Loading: 103.7% (Rating: 212 MVA)  
 DFAX: 5.4%  
 Pre-project loading: 100.6%
7. Facility: Sideburn – Yavapai 138.0 kV line  
 Contingency: Colonial Substation – Idaho Springs 138.0 kV  
 Loading: 103.1% (Rating: 292 MVA)  
 DFAX: 16.3%  
 Pre-project loading: 96.3%

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                   | Required Upgrade                            | Cost Estimate | Allocation to Project |
|--------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Rocky Reach – Idaho Springs 138.0 kV       | Rebuild line with higher-capacity conductor | \$1,303,000   | 19% (\$250,000)       |
| Centennial – Glen Carlyn 138.0 kV line     | Reconductor line                            | \$440,000     | 8% (\$33,000)         |
| Rocky Reach – Burke-Sideburn 138.0 kV      | Reconductor line                            | \$452,000     | 19% (\$87,000)        |
| Annandale Substation – Centennial 138.0 kV | Reconductor line                            | \$446,000     | 8% (\$34,000)         |
| Gallows Road – Idylwood 138.0 kV line      | Reconductor line                            | \$431,000     | 8% (\$36,000)         |
| Covanta Fairfax Energy – Occoquan 138.0 kV | Reconductor line                            | \$1,092,000   | 6% (\$69,000)         |
| Yavapai – Sideburn 138.0 kV                | Reconductor line                            | \$590,000     | 25% (\$146,000)       |
|                                            | Total allocated to project                  |               | \$655,000             |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.